OR4023 (Optimization) Homework 4

Name: _____ Student ID: _____

Question 1. Let

$$f(x_1, x_2) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1^3 + x_1^4.$$

Suppose that the function is minimized starting from $\mathbf{x}_0 = (0, -2)^T$. Verify that $\mathbf{p}_0 = (0, 1)^T$ is a direction of descent.

Question 2. Consider the quadratic function

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} - \mathbf{c}^T\mathbf{x},$$

where $\mathbf{A}$ is a positive-definite matrix. Let $\mathbf{p}$ be a direction of descent for f at the point $\mathbf{x}$. Prove that the solution of the exact line search problem

$$\min_{\alpha>0} f(\mathbf{x} + \alpha\mathbf{p})$$

is

$$\alpha = -\frac{\nabla f(\mathbf{x})\mathbf{p}}{\mathbf{p}^T\mathbf{A}\mathbf{p}}.$$

Question 3. Consider the problem

$$\text{minimize } f(x_1, x_2) = x_1^2 + 2x_2^2.$$

(a) If the starting point is $\mathbf{x}_0 = (2, 1)^T$, show that the sequence of points generated by the steepest-descent algorithm is given by

$$\mathbf{x}_k = \left(\frac{1}{3}\right)^k \begin{pmatrix} 2 \\ (-1)^k \end{pmatrix}$$

if an exact line search is used.

(b) Show that $f(\mathbf{x}_{k+1}) = f(\mathbf{x}_k)/9$.

(c) Compare the results in (b) to the bounds on the convergence rate of the steepest descent method when minimizing a quadratic function. What conclusions can you draw regarding this method?

Question 4. Apply Newton's method to solve $f(x) = x^2 - a = 0$, where $a > 0$. This is a good way to compute $\pm\sqrt{a}$. How does the iteration behave if $a \leq 0$?

Question 5. Consider the problem

$$\min f(x_1, x_2) = (x_1 - 2x_2)^2 + x_1^4.$$

(a) Apply the steepest descent method with initial guess $(2, 1)^T$ for 2 iterations.

(b) Apply the classical Newton's method with initial guess $(2, 1)^T$ for 2 iterations.